Parent–Offspring Conflict & Sibling Competition in Big Brown Bats

Tang Tang Guo, Iman Nemar

QMEE Presentation

Parent–Offspring Conflict & Sibling Competition

<u>Parent–Offspring Conflict:</u>

A disagreement between parents and offspring about investment of parental resources

<u>Sibling Competition:</u>

Competition between siblings for resources such as food

Bats!

- 1500 species of bats worldwide
- Long pregnancies of 2-7 months
- Give birth to 1-4 pups a year

Big Brown Bats

Give birth to either:

- one pup
  (in western North America)
- two pups
  (in eastern North America)

Research Question #1

How do parental care behaviours change over pup development?

If moms invest less in parental care over time, we predict to see…

- Decreased successful attachment attempts to teat
- Decreased guidance of pups towards teat

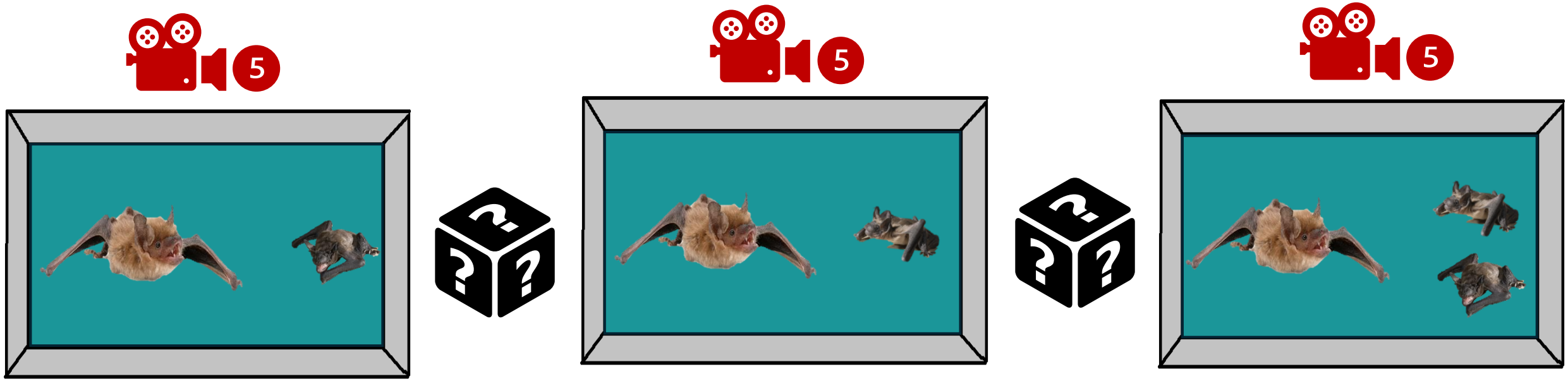
Research Question #2

How do pup behaviours change in the presence/absence of a competitor (i.e. their sibling)?

If there is sibling competition, then in the presence of a sibling, we predict to see…

- Decreased successful attachment attempts to teat
- Decreased guidance of each pup

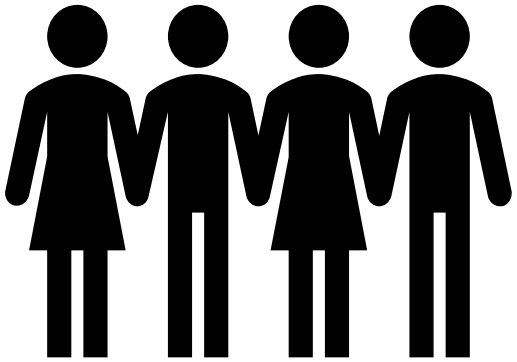
Experimental Design



Video Scoring & ICC Scores

Intraclass Correlation Coefficient (ICC):

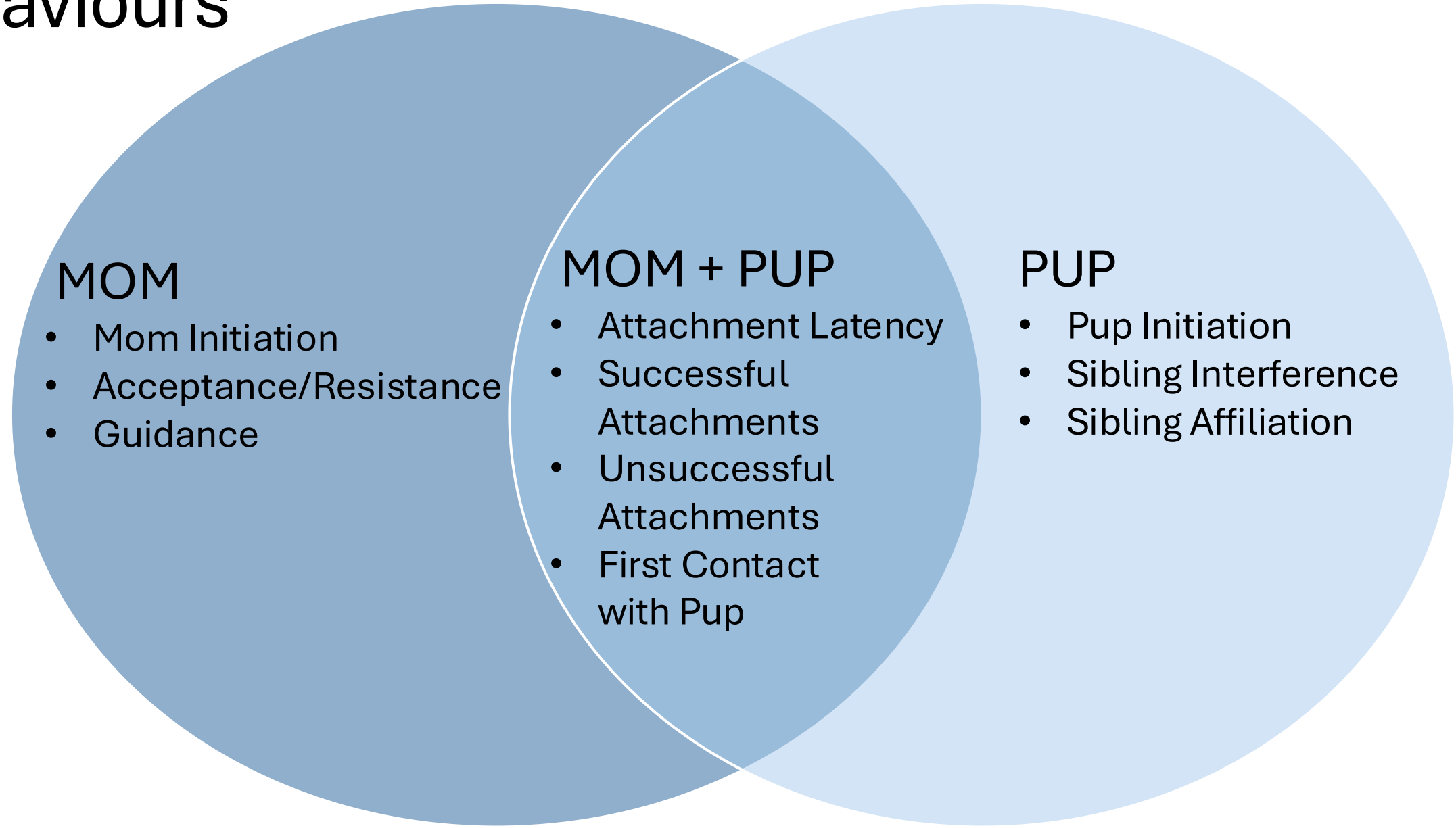
Reliability index in test-retest, intra-rater, and **inter-rater reliability** analyses.



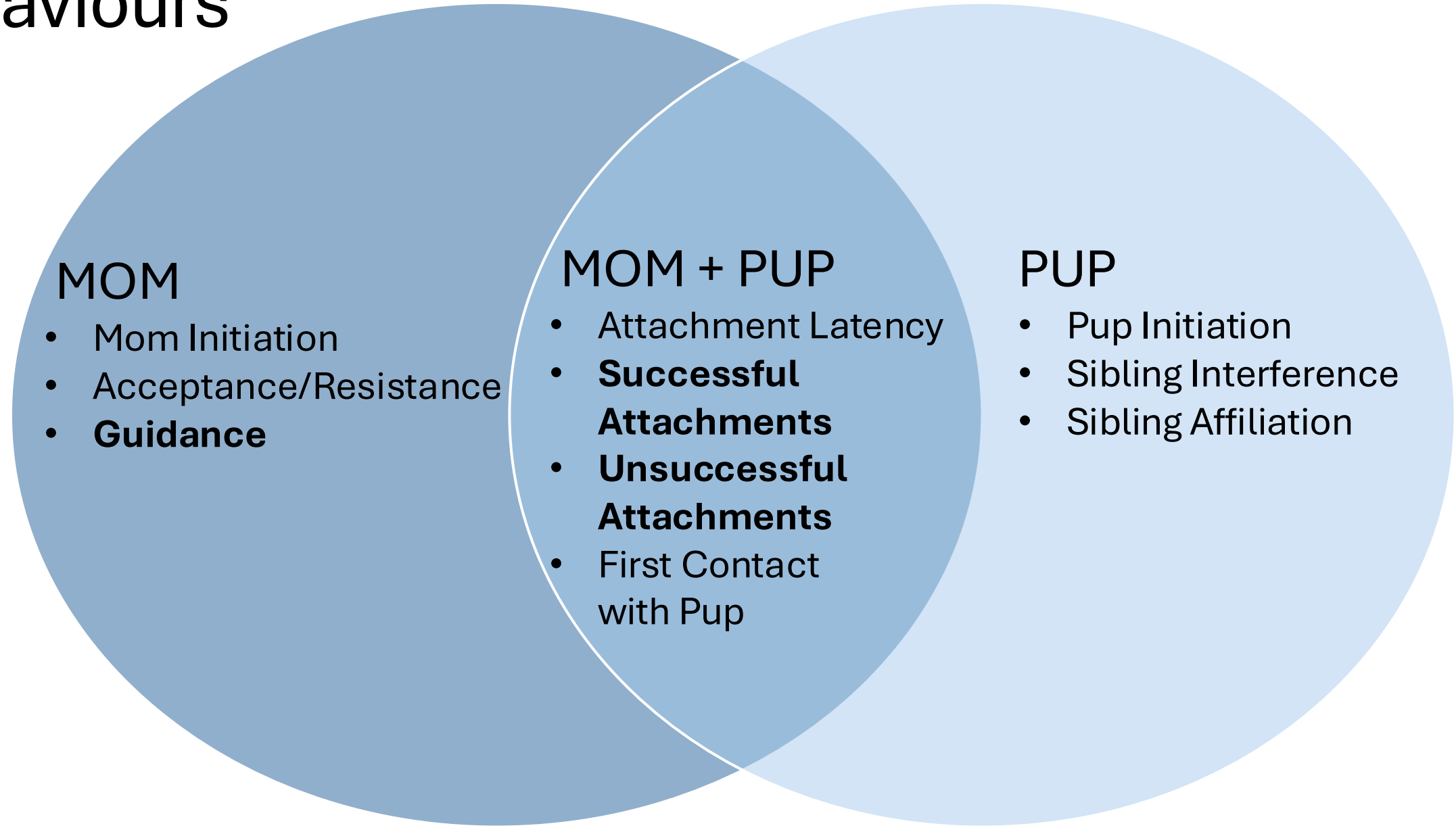
Army of Raters

ICC Score Range	Behaviours
≥ 0.9	• Let Go • First Attachment • Successful Attachment (frequency)
0.8 < 0.9	• Unsuccessful Attachment (frequency) • Guidance (frequency + duration) • Sibling Cooperation (frequency) • Sibling Interference (frequency)
0.7< 0.8	• Mom/Pup Initiation • Sibling Interference (duration)
≤0.7	• Acceptance/Resistance • Sibling Cooperation (duration)

Behaviours


MOM
Mom Initiation
Acceptance/Resistance
Guidance
MOM + PUP
Attachment Latency
Successful Attachments
Unsuccessful Attachments
First Contact with Pup
PUP
Pup Initiation
Sibling Interference
Sibling Affiliation

Behaviours


MOM
Mom Initiation
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MOM + PUP
Attachment Latency
Successful Attachments
Unsuccessful Attachments
First Contact with Pup
PUP
Pup Initiation
Sibling Interference
Sibling Affiliation

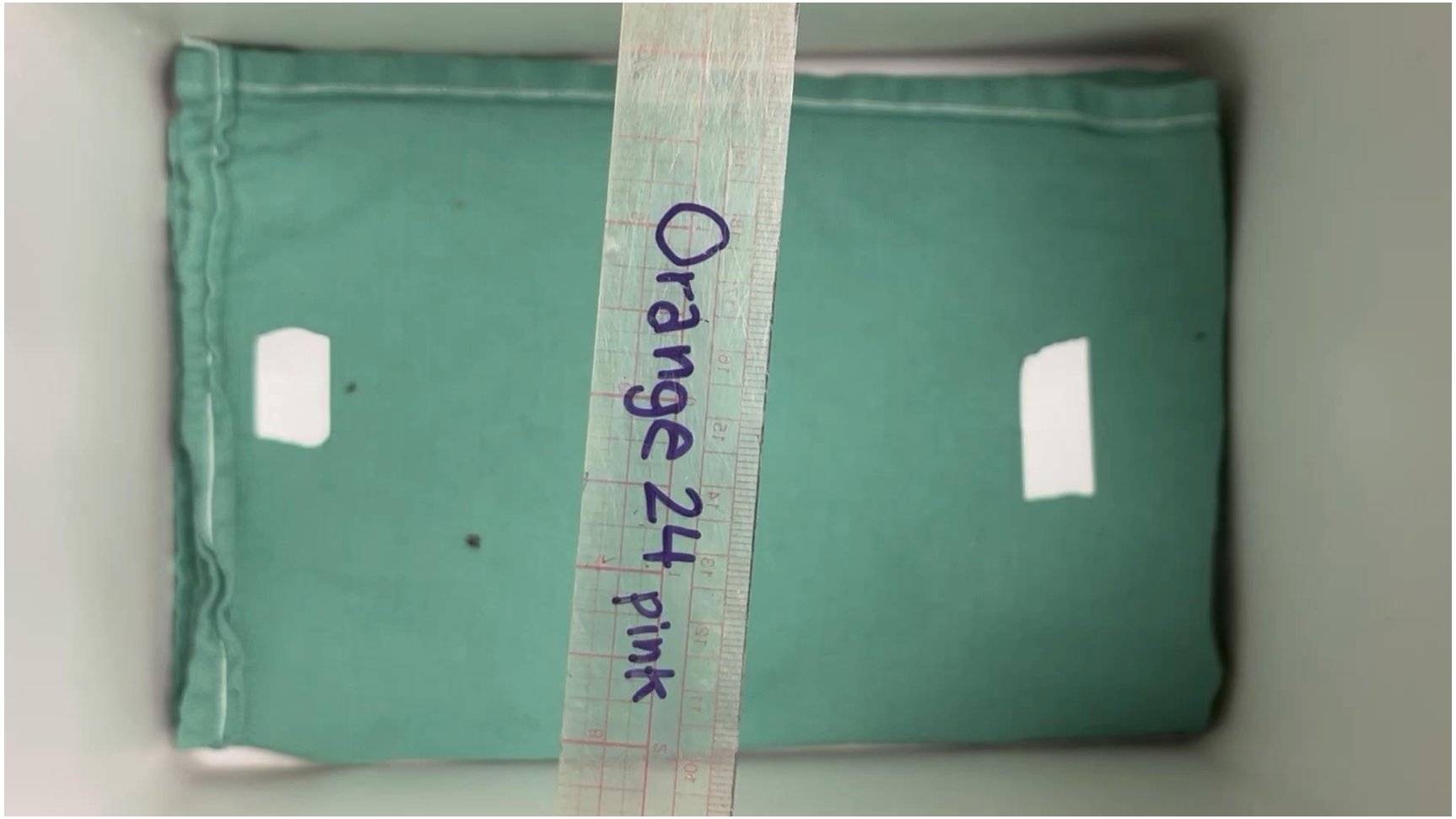
Dataset

	Rater	Condition	PND	ObservationID	MomID	Behaviour	BehaviourType	Start_s	Stop_s	Duration_s	PupID
67	Easton	Double	2	Grey112_both_26June2025	Grey112	MOM INITIATION	POINT	43.844	43.844	0.000	Grey112_pink
68	Easton	Double	2	Grey112_both_26June2025	Grey112	FIRST ATTACHMENT	POINT	77.177	77.177	0.000	Grey112_pink
69	Easton	Double	2	Grey112_both_26June2025	Grey112	ACCEPTANCE	POINT	348.843	348.843	0.000	Grey112_pink
70	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	FIRST TOUCH	POINT	14.967	14.967	0.000	Orange07_blue
71	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	GUIDANCE	STATE	18.400	35.833	17.433	Orange07_blue
72	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	GUIDANCE	STATE	42.667	61.335	18.668	Orange07_blue
73	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	GUIDANCE	STATE	97.968	115.368	17.400	Orange07_blue
74	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	SUCCESSFUL ATTACHMENT	STATE	61.335	67.668	6.333	Orange07_blue
75	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	UNSUCCESSFUL ATTACHMENT	POINT	35.833	35.833	0.000	Orange07_blue
76	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	MOM INITIATION	POINT	14.967	14.967	0.000	Orange07_blue
77	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	FIRST ATTACHMENT	POINT	61.335	61.335	0.000	Orange07_blue
78	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	OBSTRUCTION	STATE	183.537	244.234	60.697	Orange07_blue
79	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	ACCEPTANCE	POINT	244.234	244.234	0.000	Orange07_blue
80	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	FIRST TOUCH	NA	NA	NA	0.000	Orange07_orange
81	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	GUIDANCE	STATE	73.868	83.302	9.434	Orange07_orange
82	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	SUCCESSFUL ATTACHMENT	STATE	88.168	183.537	95.369	Orange07_orange
83	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	UNSUCCESSFUL ATTACHMENT	NA	NA	NA	0.000	Orange07_orange
84	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	MOM INITIATION	POINT	69.302	69.302	0.000	Orange07_orange
85	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	FIRST ATTACHMENT	POINT	88.168	88.168	0.000	Orange07_orange
86	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	OBSTRUCTION	STATE	183.537	244.234	60.697	Orange07_orange
87	Easton	Double	2	Orange07_BothPups_11June2024	Orange07	ACCEPTANCE	POINT	244.234	244.234	0.000	Orange07_orange
88	Easton	Double	2	Orange14_BothPups_10June2024	Orange14	FIRST TOUCH	POINT	13.667	13.667	0.000	Orange14_pink

Dataset

```
   Rater       Condition         PND                                ObservationID
Easton:1922   Double:3947   Min.   : 2.00   Orange22_BothPups_28June2024:  81
Ella  : 943   Single:1873   1st Qu.: 6.00   Orange21_BothPups_01July2024:  69
Emily :1957                 Median :10.00   Orange07_BothPups_11June2024:  66
Kevin : 998                 Mean   :13.19   Orange19_BothPups_09July2024:  66
                            3rd Qu.:22.00   Orange19_BothPups_23June2024:  66
                            Max.   :30.00   Orange23_BothPups_03July2024:  65
                                            (Other)                     :5407
     MomID                          Behaviour    BehaviourType    Start_s
Orange07: 451   UNSUCCESSFUL ATTACHMENT: 954   POINT:3044    Min.   :  2.035
Orange27: 450   GUIDANCE                : 845   STATE:1269    1st Qu.: 21.595
Orange29: 449   SUCCESSFUL ATTACHMENT   : 690   NA's :1507    Median : 50.533
Orange21: 445   LET GO                  : 453                 Mean   : 93.855
Orange19: 432   FIRST TOUCH             : 434                 3rd Qu.:130.268
Orange14: 421   SIBLING COOPERATION     : 397                 Max.   :381.549
(Other) :3172   (Other)                 :2047                 NA's   :1507
    Stop_s          Duration_s                 PupID
Min.   :  2.035   Min.   :  0.00   105863_pink    : 194
1st Qu.: 23.867   1st Qu.:  0.00   Orange29_blue  : 189
Median : 63.670   Median :  0.00   Orange07_blue  : 188
Mean   :111.655   Mean   : 13.73   Orange07_orange: 187
3rd Qu.:179.703   3rd Qu.:  0.00   Orange27_nc    : 187
Max.   :381.549   Max.   :300.00   (Other)        :3901
```

Frequency + Duration of Guidance



The Approach (Frequency Data)

What kind of model are we going to make?

- Ratio (frequency, duration) data
- Created a GLM with negative binomial to account for overdispersion

```
GuidanceModel_00 <- glmmTMB(
  freq ~ Condition * PND,
    data = Guidance2,
  family = nbinom2)
```

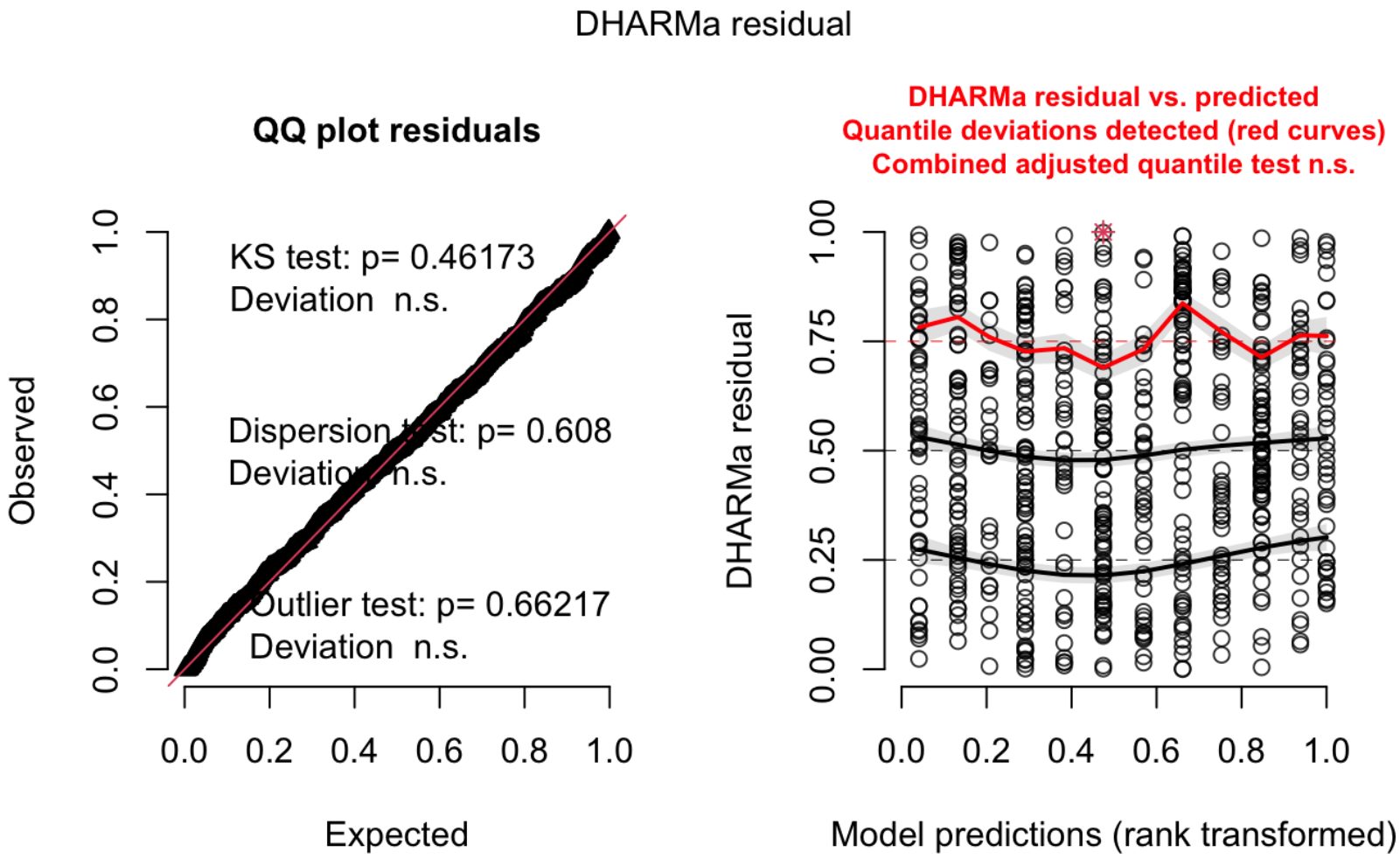
What fixed/random effects should we be accounting for?

- Fixed Effects:
  - Condition (single vs double)
  - PND (age)
- Random Effects:
  - MomID & Pup ID

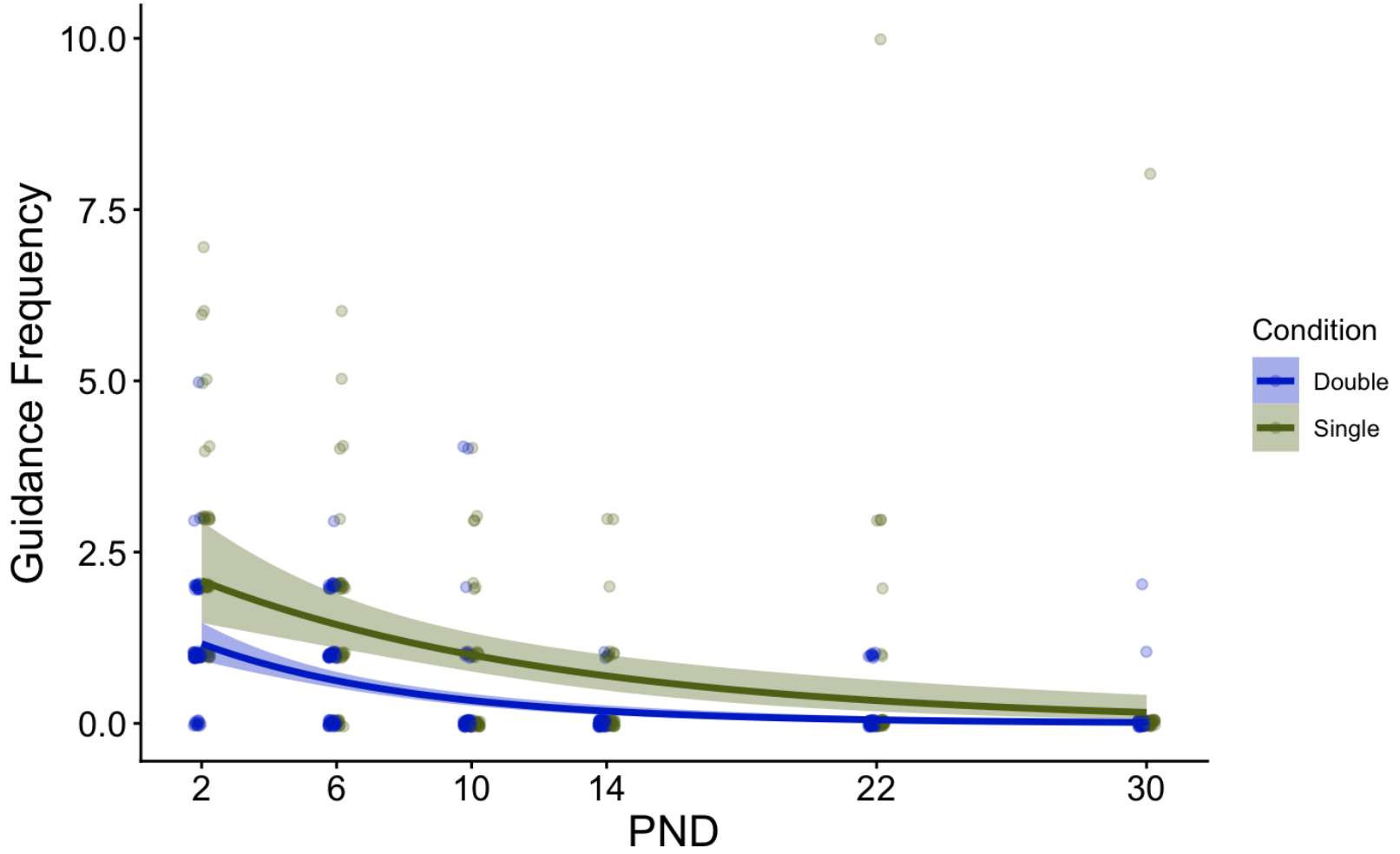
```
GuidanceModel_02 <- glmmTMB(
  freq ~ Condition * PND +
    (1+PND|MomID) +
    (1+Condition|PupID),
    data = Guidance2,
  family = nbinom2)
```

Negative binomial vs. beta binomial vs. quasi binomial?

Guidance (Frequency) DHARMa



Pups receive more frequent guidance when alone than with their sibling



The Approach (Duration Data)

What kind of model are we going to make?

- Continuous, ratio data, which include zero values
- Created 2 GLMs using the **hurdle model**
  - **Binomial GLM** – to model zero/non-zero data
  - **Gamma GLM** – to model continuous non-zero values

What fixed effects/random effects should we be accounting for?

- Fixed Effects:
  - Condition (single vs double)
  - PND (age)
- Random Effects:
  - MomID & Pup ID

Should we include rater effect in all of our models or only the ones with poor ICC scores?

The Approach (Duration Data)

What kind of model are we going to make?

```
GuidanceDur_binom_00 <- glm(
  dur > 0 ~ Condition * PND,
  family = binomial(link = "logit"),
  data = Guidance2)
```

```
GuidanceDur_binom <- glmer(
  dur > 0 ~ Condition * PND +
    (1+PND | MomID) +
    (1+Condition | PupID),
  family = binomial(link = "logit"),
  data = Guidance2)
```

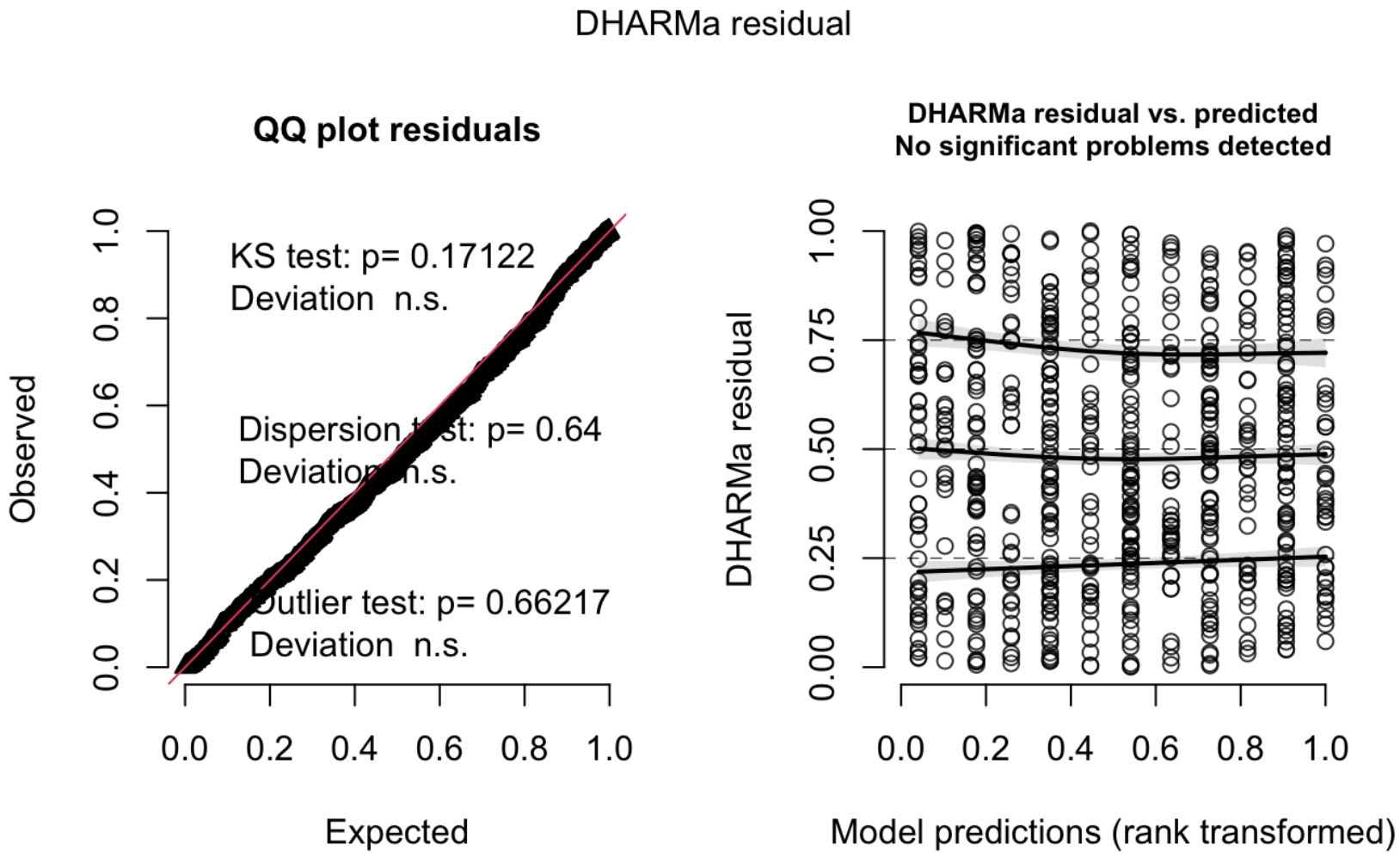
What fixed effects/random effects should we be accounting for?

```
GuidanceDur_gamma_00 <- glm(
  dur ~ Condition * PND,
  family = Gamma(link = "log"),
  data = GuidanceDuration_pos)
```

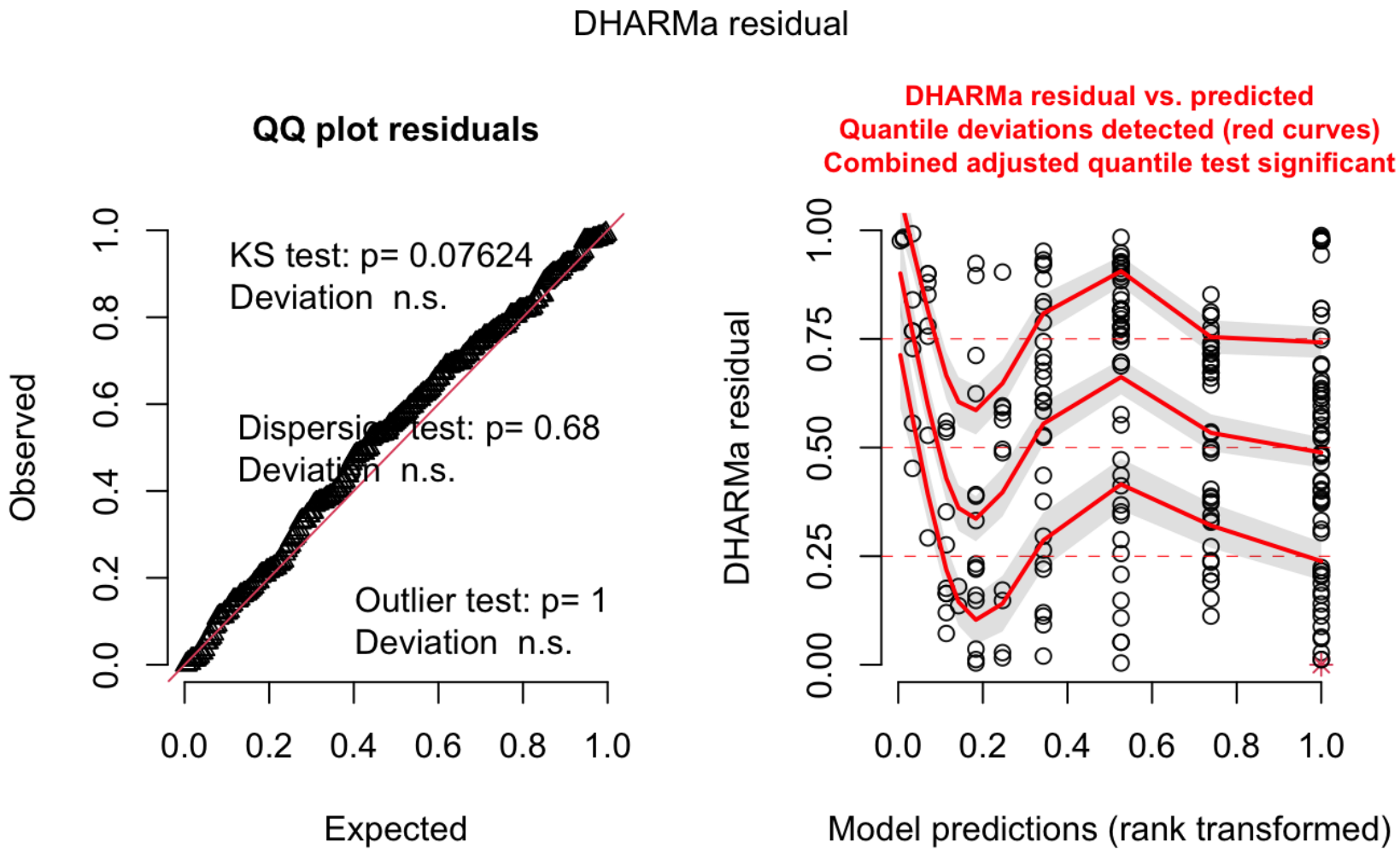
```
GuidanceDur_gamma <- glmmTMB(
  dur ~ Condition * PND +
    (1+PND | MomID) +
    (1+Condition | PupID) +
    (1|Rater),
  family = Gamma(link = "log"),
  data = GuidanceDuration_pos)
```

Should we include rater effect in all of our models or only the ones with poor ICC scores?

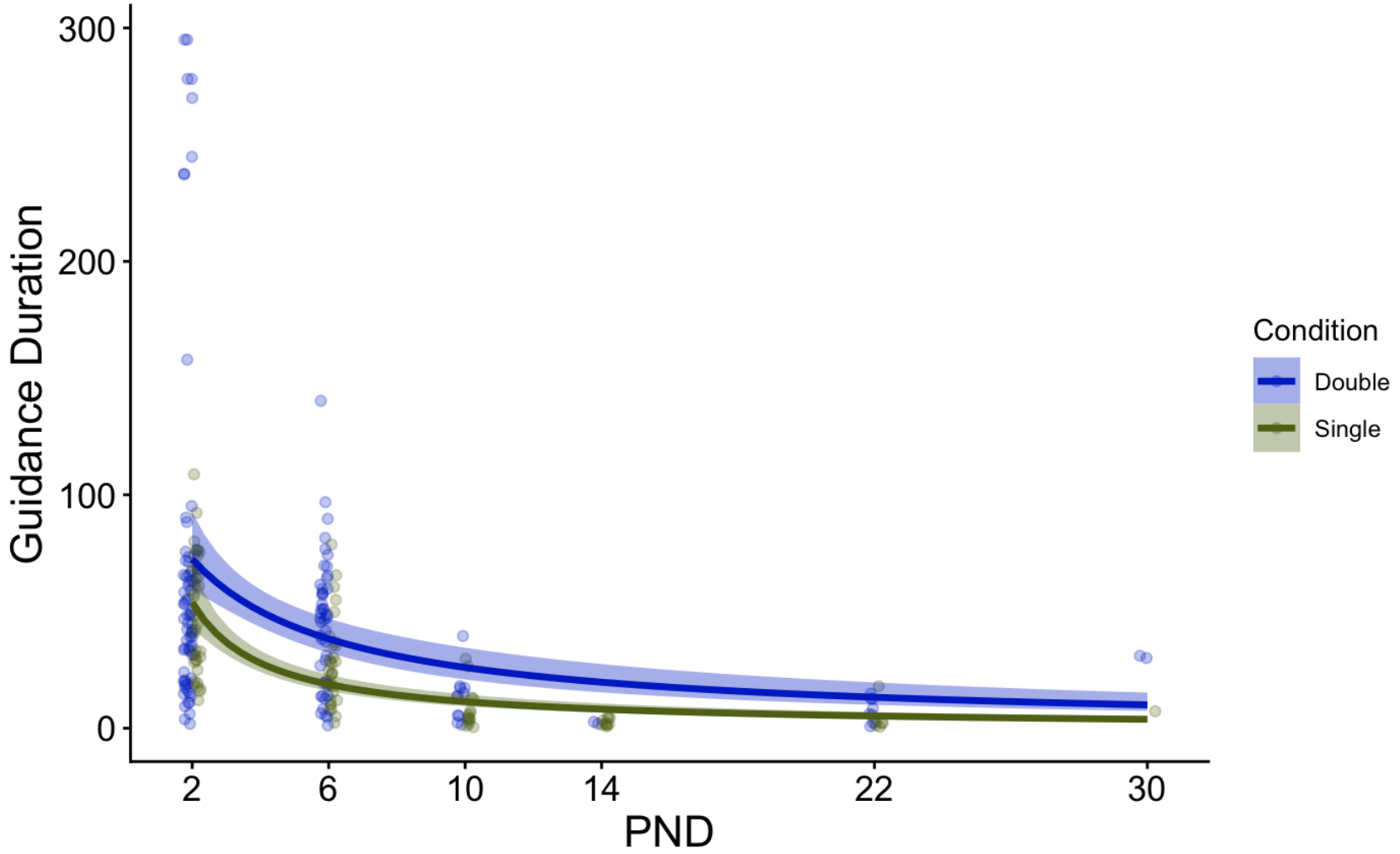
Guidance Duration DHARMa 1/2



Guidance Duration DHARMa 2/2



Pups each receive longer guidance in the presence of their sibling



Successful Attachments

The approach (successful attachments data)

What kind of model are we going to make?

- Created GLM of proportions of successful to total # of attachments

```
PropSuccessModel_00 <- glmmTMB(
  cbind(`SUCCESSFUL ATTACHMENT`,
        `UNSUCCESSFUL ATTACHMENT`) ~ Condition * PND,
    data = Success_Unsuccess_Wide2,
  family = binomial(link = "logit"))
```

What fixed effects/random effects should we be accounting for?

Fixed Effects of:

Condition (single vs double)

PND (age)

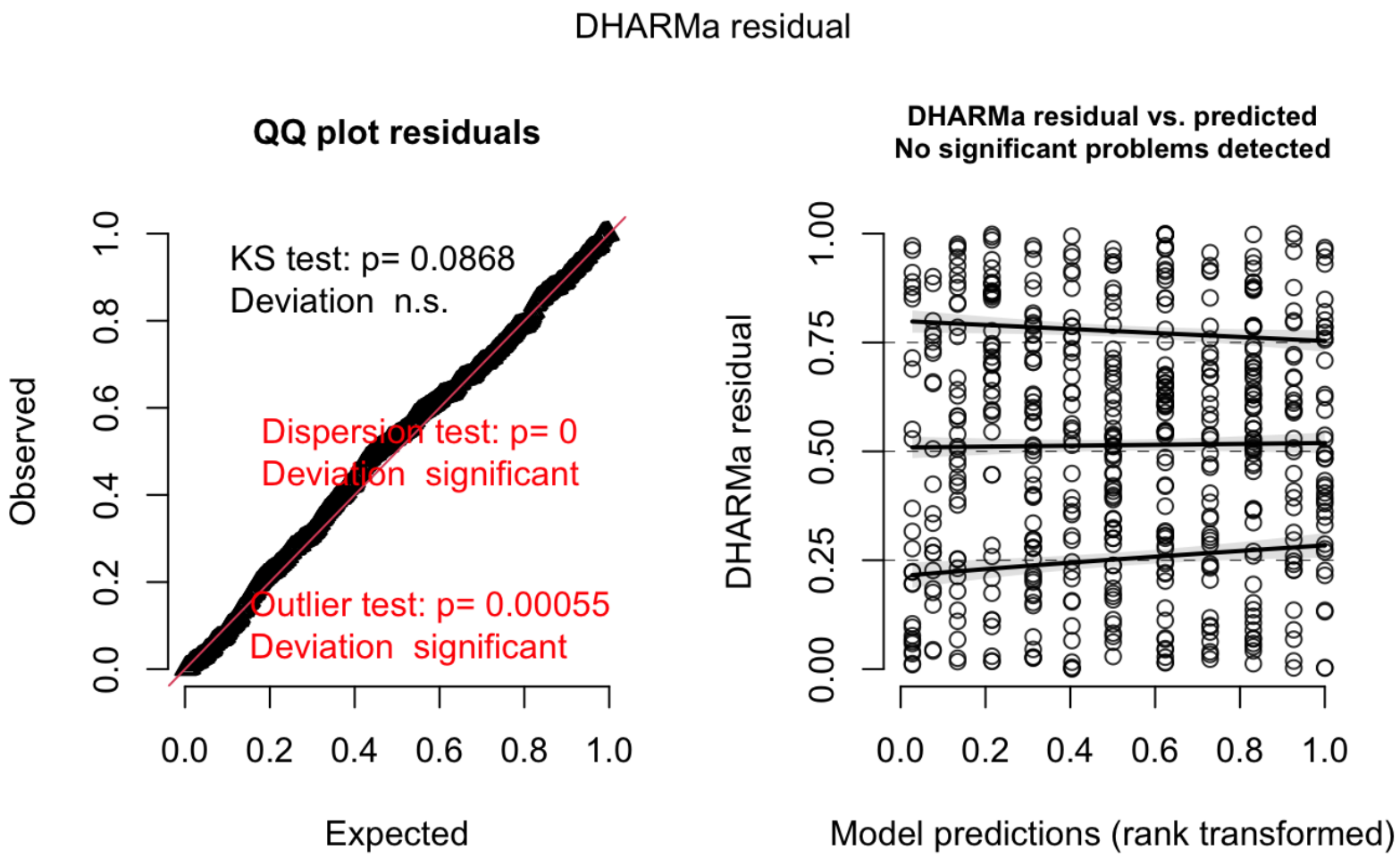
Random Effects of:

MomID & Pup ID

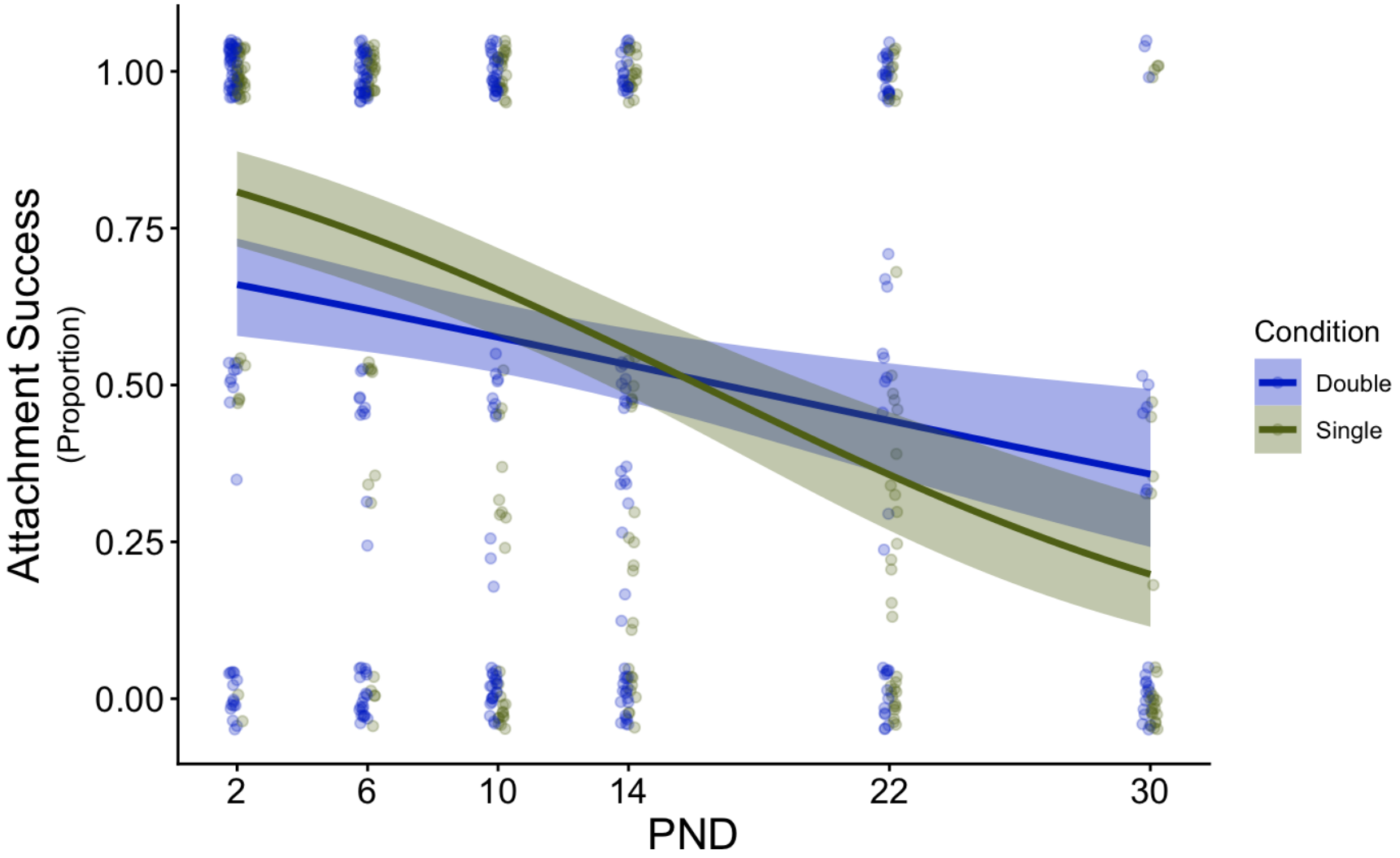
```
PropSuccessModel_02 <- glmmTMB(
  cbind(`SUCCESSFUL ATTACHMENT`,
        `UNSUCCESSFUL ATTACHMENT`) ~ Condition * PND +
    (1+Condition|PupID) +
    (1| MomID) +
    (1| ObservationID),
    data = Success_Unsuccess_Wide2,
  family = binomial(link = "logit"))
```

What are observation level random effects and how do we account for them?

Attachment Success DHARMa



Attachment success is higher for single pups early in development



Next Steps

- Finish creating the models for all behaviours
- Create visualizations for models
- Survive!